Chapter 13 — I/O and NIO

Almost every real program reads or writes something outside itself: configuration files, CSV exports (CSV = comma-separated values, a plain-text table with one row per line), logs, uploaded images, the keyboard, the terminal. **I/O** (input/output) is how a program moves data across that boundary. Java has two generations of file APIs that you will meet side by side in real code: the original `java.io` package (Java 1.0, built around *streams* of bytes and characters) and **NIO.2** (`java.nio.file`, Java 7, built around `Path` and the `Files` utility class). This chapter teaches both, shows which one to reach for, and covers the topics that trip people up in production: buffering, character encodings (the rules for turning text into bytes), serialization (turning objects into bytes so they can be saved or sent), and configuration files.

You already know the tools this chapter leans on: try-with-resources (the `try (Resource r = …) { }` form that closes `r` automatically, even after an exception) and `IOException` from Chapter 8, and lazy streams from Chapter 12 (`Files.lines` and `Files.walk` return `Stream`s).

What you'll learn

§	Topic	You'll be able to…
13.1	The big picture	tell `java.io` and NIO.2 apart and pick the right API
13.2	`Path`	build, combine, normalise and compare file paths without touching the disk
13.3	`Files`: managing files and directories	create, copy, move, delete, inspect, list and walk directory trees safely
13.4	Reading and writing whole files	use the `Files` one-liners and stream large files line by line
13.5	Byte streams and character streams	understand `InputStream` / `OutputStream` / `Reader` / `Writer` and the decorator chain
13.6	Buffering	know why unbuffered I/O is slow and why forgetting `flush` loses data
13.7	Charsets and encodings	avoid mojibake (garbled text, defined in §13.7), and know what changed with UTF-8 by default (Java 18)
13.8	Console and standard streams	read user input with `Scanner`, `BufferedReader`, `Console` and `IO`
13.9	Binary data, channels and buffers	read/write primitives, use `FileChannel` and `ByteBuffer`, know what `WatchService` is for
13.10	Serialization (and why to avoid it)	explain how it works, `transient`, `serialVersionUID`, filters, and its security risk
13.11	Properties files	load, read and store `key=value` configuration
13.12	Working with JSON	know that JSON needs a library, and what the code looks like
13.13	Best practices and common mistakes	a checklist for correct, portable I/O

All runnable code is in `code/` (package `ch13.*`), compiled with `javac --release 25` (the Java compiler, targeting Java 25) and run on JDK 26 (the Java Development Kit). Every program works inside a directory made by `Files.createTempDirectory` and deletes it at the end, so nothing is left on your disk. Outputs are pasted from real runs.

> **Exam depth.** `File` vs `Path` method-name differences, `resolve`/`relativize` puzzles, `Files` exception rules, stream class tables and `Console` details: OCP Chapter 14 — I/O. This chapter stays at "use it correctly at work" depth and links there for traps.

13.1 The big picture: `java.io` vs NIO.2

What it is

- **`java.io`** (Java 1.0) models I/O as **streams**: a one-way flow of data you read from (an *input stream*) or write to (an *output stream*), one piece after another, like water through a pipe. Files are represented by the old `java.io.File` class. (These I/O streams are unrelated to the `java.util.stream.Stream` of Chapter 12 — same word, different idea.)
- **NIO** ("New I/O", Java 1.4, `java.nio`) added **buffers** (`ByteBuffer`) and **channels** (`FileChannel`, `SocketChannel`) — lower-level, block-oriented (moving data in chunks rather than one byte at a time), and capable of **non-blocking** network I/O (a read or write returns immediately even if no data is ready, instead of making the thread wait).
- **NIO.2** (Java 7, `java.nio.file`) replaced `File` with the **`Path`** interface and the **`Files`** class full of static helpers, plus file attributes, **symbolic links** (symlinks: small files that just point to another file or directory, like a shortcut), directory walking and file-change watching.

Analogy: `java.io` is the plumbing — pipes you connect together. NIO.2 is the building manager's office — it knows where every room (file) is, can create, move or inspect rooms, and hands you a pipe when you need one.

Why it matters

Both APIs are current and interoperate. Modern code uses `Path` + `Files` for everything about *files*, and still uses `java.io` stream classes whenever data flows from something that isn't a file (network **sockets** — the OS endpoint for one network connection — HTTP bodies, the data part of a web request or response, zip entries, in-memory arrays) — `Files.newInputStream(path)` even *returns* a `java.io.InputStream`.

Need	Use	Since
Represent a file/directory location	`Path` (`Path.of(...)`)	7 (`Path.of` 11)
Create, copy, move, delete, check, list	`Files.*` static methods	7
Read/write a small text file in one call	`Files.readString` / `Files.writeString`	11
Read a huge text file line by line	`Files.lines` (lazy `Stream`) or `Files.newBufferedReader`	8 / 7
Process bytes from any source (socket, zip, HTTP)	`InputStream` / `OutputStream`	1.0
Process text from any source	`Reader` / `Writer` (+ a charset)	1.1
Random access (jumping to any position), big files, memory mapping	`FileChannel` + `ByteBuffer`	1.4
Legacy (older, still-in-use) APIs that want a `File`	`path.toFile()` / `file.toPath()`	7

Why `File` was replaced

The old `File` class has real flaws that NIO.2 fixes:

1. **Silent failures.** `file.delete()`, `mkdir()` and `renameTo()` return `false` without saying *why* (missing? permission? not empty?). `Files.delete` throws a specific exception (`NoSuchFileException`, `DirectoryNotEmptyException`, `AccessDeniedException`).
2. **No symbolic-link support, poor metadata** (data *about* a file — size, dates, owner — rather than its contents). NIO.2 reads attributes (size, times, POSIX permissions — the Unix read/write/execute flags for owner, group and others — owner) in one call.
3. **No efficient directory walking.** `File.listFiles()` loads everything into an array; `Files.walk`/`Files.list` stream lazily.

TL;DR

Use `Path` + `Files` for files and directories; use `InputStream`/`Reader` families for data flowing from anywhere; convert with `toFile()`/`toPath()` only at legacy boundaries.

13.2 `Path`: naming files and directories

What it is

A **`Path`** is an **immutable** (can't be changed after creation) object that represents a location in a file system as a sequence of **name elements** (`reports`, `2025`, `q3`, `sales.csv`) plus an optional **root** (`/` on Unix — the family of operating systems that includes Linux and macOS — `C:\` on Windows). Crucially, a `Path` is *just a name*: creating one does not check that the file exists or touch the disk at all. Think of it as a street address written on paper — you can write down an address for a house that hasn't been built.

Key terms:

- **Absolute path** — starts at a root (`/srv/app/logs`). **Relative path** — interpreted from the **current working directory** (the folder the program was started from) of the JVM **process** (a running program with its own memory, managed by the OS) (`reports/q3.csv`).
- **Separator** — `/` on Unix/macOS, `\` on Windows. `Path.of("a", "b")` inserts the right one, so don't concatenate strings with `"/"` yourself.
- `.` means "this directory", `..` means "the parent directory".

How it works

1. Create a path: `Path.of("reports", "2025", "sales.csv")` (Java 11) or the older, identical `Paths.get(...)` (Java 7).
2. Take it apart: `getFileName()`, `getParent()`, `getRoot()`, `getNameCount()`, `getName(i)`, `subpath(from, to)`.
3. Combine paths: `resolve` (append), `resolveSibling` (replace the last element), `relativize` (the route from one path to another).
4. Clean it: `normalize()` removes `.` and resolves `name/..` pairs *textually*; `toAbsolutePath()` prefixes the working directory; `toRealPath()` asks the file system (resolves symlinks and **requires the file to exist**).

`code/ch13/paths/PathBasics.java` (key parts):

```java
Path p = Path.of("reports", "2025", "q3", "sales.csv");
p.getFileName(); p.getParent(); p.getRoot(); p.getNameCount(); p.getName(1); p.subpath(1, 3);

Path base = Path.of("/srv/app");
base.resolve("logs/app.log");                // append a relative path
base.resolve("/etc/hosts");                  // an absolute argument simply wins
base.resolve("config.yml").resolveSibling("secrets.yml");

Path.of("/srv/app/./logs/../data/./users.db").normalize();
Path from = Path.of("/srv/app/logs"), to = Path.of("/srv/app/data/users.db");
from.relativize(to);                         // how do I get from 'from' to 'to'?
to.startsWith("/srv/ap");                    // compares whole NAME elements, not characters
```

Output:

```
path          = reports/2025/q3/sales.csv
fileName      = sales.csv
parent        = reports/2025/q3
root          = null
nameCount     = 4
name(1)       = 2025
subpath(1,3)  = 2025/q3
isAbsolute    = false
resolve       = /srv/app/logs/app.log
resolve abs   = /etc/hosts
resolveSibling= /srv/app/secrets.yml
normalize     = /srv/app/data/users.db
relativize    = ../data/users.db
round trip    = /srv/app/data/users.db
startsWith    = true / string-wise 'app' = false
endsWith      = true
File <-> Path = reports/2025/q3/sales.csv -> true
Paths.get     = true
```

What to notice:

- `getRoot()` is `null` for a relative path, and `getName(0)` is the first element *after* the root.
- `subpath(1, 3)` is **half-open** (includes the start index, excludes the end) like `String.substring`: elements 1 and 2.
- `resolve` with an absolute argument returns the argument — a classic source of bugs when a user-supplied "file name" starts with `/`.
- `relativize` requires both paths to be of the same kind (both absolute or both relative), otherwise it throws `IllegalArgumentException`.
- `startsWith("/srv/ap")` is `false`: comparison is by name elements, not by characters.

Common mistakes

- Building paths by string concatenation (`dir + "/" + name`) — breaks on Windows and doubles separators. Use `resolve`.
- Assuming `normalize()` checks the disk. It is purely textual; `a/link/..` normalises to `a` even if `link` is a symlink pointing elsewhere. Use `toRealPath()` when it matters.
- **Path traversal**: `uploads.resolve(userInput)` where `userInput` is `../../etc/passwd`. Guard with `Path target = uploads.resolve(name).normalize(); if (!target.startsWith(uploads)) throw ...` — this is exactly the check web servers do. (Path traversal is an attack where `..` segments in input escape the folder you meant to allow.)

Self-check

1. Does `Path.of("missing.txt")` throw if the file doesn't exist? → No; a `Path` is only a name.
2. `Path.of("/a/b").resolve("/c")`? → `/c`.
3. `Path.of("/a/b").relativize(Path.of("/a/c/d"))`? → `../c/d`.

13.3 `Files`: managing files and directories

What it is

`java.nio.file.Files` is a class of static methods that *do* touch the disk: test, create, copy, move, delete, read attributes, list and walk. Every method that can fail throws a specific subclass of `IOException` (a *checked* exception, so the compiler makes you handle it — Chapter 8 §8.3).

How it works — the core operations

Task	Method	Fails with (if…)
Does it exist?	`exists`, `notExists`, `isRegularFile`, `isDirectory`, `isReadable`, `isSymbolicLink`	never throws; returns `false` if unsure
Make a directory	`createDirectory(p)` / `createDirectories(p)`	`FileAlreadyExistsException` / parents made too, no error if present
Make an empty file	`createFile(p)`	`FileAlreadyExistsException`
Temporary file/dir	`createTempFile(prefix, suffix)`, `createTempDirectory(prefix)`	in `java.io.tmpdir` (the system's temp folder)
Copy	`copy(src, dst, options...)`	`FileAlreadyExistsException` unless `REPLACE_EXISTING`
Move / rename	`move(src, dst, options...)`	same; add `ATOMIC_MOVE` for all-or-nothing (**atomic**: other programs see either the old state or the new one, never a half-done step) within one file system
Delete	`delete(p)` / `deleteIfExists(p)`	`NoSuchFileException`, `DirectoryNotEmptyException` / returns `boolean`
Size, times, type	`size`, `getLastModifiedTime`, `readAttributes(p, BasicFileAttributes.class)`	`NoSuchFileException`
Same content?	`mismatch(a, b)` (Java 12) → `-1` if identical, else first differing byte index	
Same file?	`isSameFile(a, b)` — follows links	
One level listing	`list(dir)` → `Stream<Path>` (lazy, **close it**)	`NotDirectoryException`
Recursive walk	`walk(dir[, maxDepth])`, `find(dir, maxDepth, (path, attrs) -> …)` → `Stream<Path>`	
Fine-grained walk with callbacks	`walkFileTree(dir, FileVisitor)`	

Details worth knowing:

- `copy` of a directory copies only the directory itself (empty), not its contents. Recursive copy = `walk` + `copy` each entry.
- `walk` visits **depth-first** (goes all the way down one branch of the folder tree before moving to the next), starts with the directory itself, and by default does **not** follow symbolic links (pass `FileVisitOption.FOLLOW_LINKS` to change that).
- `readAttributes(..., BasicFileAttributes.class)` fetches size, times and type in **one** file-system call — much cheaper than calling `size`, `isDirectory` and `getLastModifiedTime` separately in a loop over thousands of files.

`code/ch13/files/FileOps.java` (key parts):

```java
Path root = Files.createTempDirectory("java1315_fileops");
try {
    Path docs = Files.createDirectories(root.resolve("docs/2025"));   // makes parents too
    Path a = Files.writeString(docs.resolve("a.txt"), "alpha\n");
    Files.createDirectory(docs);                                       // ->
FileAlreadyExistsException
    Path b = Files.copy(a, docs.resolve("b.txt"));
    Files.copy(a, b);                                                  // ->
FileAlreadyExistsException
    Files.copy(a, b, StandardCopyOption.REPLACE_EXISTING);            // allowed
    Path moved = Files.move(b, archive.resolve("b-old.txt"));         // move = rename
    Files.mismatch(a, moved) == -1;                                    // identical content
    BasicFileAttributes attrs = Files.readAttributes(a, BasicFileAttributes.class);

    try (Stream<Path> s = Files.walk(root)) {                          // lazy + holds OS handles
        s.map(p -> root.relativize(p).toString()).sorted().forEach(...);
    }
    try (Stream<Path> s = Files.find(root, 10,
            (p, at) -> at.isRegularFile() && p.toString().endsWith(".md"))) { ... }
    Files.delete(docs);                                                // ->
DirectoryNotEmptyException
} finally {
    deleteRecursively(root);
}

static void deleteRecursively(Path root) throws IOException {
    try (Stream<Path> s = Files.walk(root)) {
        List<Path> all = s.sorted(Comparator.reverseOrder()).toList();  // children before parents
        for (Path p : all) Files.delete(p);
    }
}
```

Output:

```
exists a.txt?        true
isDirectory docs?    true
size a.txt           6 bytes
createDirectory again -> FileAlreadyExistsException
copy onto existing   -> FileAlreadyExistsException
after move b exists? false, moved exists? true
same content a/moved true
attrs: regular=true dir=false size=6 created<=modified true
-- Files.list(root) (one level, sorted):
   archive
   docs
   readme.md
-- Files.walk(root) (recursive, sorted):
   ''
   'archive'
   'archive/b-old.txt'
   'docs'
   'docs/2025'
   'docs/2025/a.txt'
   'docs/2025/notes.md'
   'readme.md'
-- Files.find *.md:
   docs/2025/notes.md
   readme.md
delete non-empty dir -> DirectoryNotEmptyException
deleteIfExists(missing) = false
read missing file    -> NoSuchFileException
temp dir removed?    true
```

- The empty string `''` in the walk output is the root itself (`root.relativize(root)`), proving that `walk` includes its starting point.
- We **sort** listing output because the order returned by the operating system is not specified; never rely on it.
- `deleteRecursively` sorts paths in reverse so `docs/2025/a.txt` comes before `docs/2025`, which comes before `docs` — a directory can only be deleted once it's empty. (`walkFileTree` with a `postVisitDirectory` callback is the other standard way.)

Watching a directory for changes

`WatchService` (Java 7) lets a program be notified when files in a directory are created, modified or deleted — how IDEs (Integrated Development Environments such as IntelliJ IDEA or VS Code) and dev servers notice edits:

```
// (not run: event delivery timing is OS-dependent; macOS uses polling with a delay of seconds)
try (WatchService ws = FileSystems.getDefault().newWatchService()) {
    dir.register(ws, StandardWatchEventKinds.ENTRY_CREATE, StandardWatchEventKinds.ENTRY_MODIFY);
    WatchKey key = ws.take();                          // blocks until something happens
    for (WatchEvent<?> e : key.pollEvents()) System.out.println(e.kind() + " " + e.context());
    key.reset();                                       // re-arm, or no more events arrive
}
```

It watches a single directory (not subdirectories), and events can be coalesced (merged into one) or overflow (`OVERFLOW` kind), so treat events as hints and re-scan when in doubt.

Common mistakes

- **Not closing** `Files.list` / `walk` / `find` / `lines`. They hold an open directory or **file handle** (an OS-issued token for an open file; each process may hold only a limited number); in a long-running server, leaking them eventually hits "Too many open files". Always use try-with-resources.

- **Check-then-act**: `if (!Files.exists(p)) Files.createFile(p);` — another process can create it in between (a **race condition**: the result depends on the timing of two independent actors). Call `createFile` and catch `FileAlreadyExistsException` instead (or use `CREATE_NEW`, §13.4).
- Using `Files.exists` as proof: it returns `false` both for "doesn't exist" and "can't tell" (no permission). `Files.notExists` is not simply its negation — both can be `false`.
- Hard-coding `/tmp` — use `Files.createTempDirectory` (portable, unique name).

Real-world examples

Log rotation (periodically renaming the current log file and starting a fresh one; `move` with `ATOMIC_MOVE` so readers never see half-written files), build tools walking source trees (`walk` + `find`), and "write to a temp file, then atomically move over the real file" — the standard trick for crash-safe config and **cache** (a saved copy of data kept for fast reuse) updates.

In the interview

> **30-second answer:** "I use `Path.of` and the `Files` class: `createDirectories`, `copy` / `move` with explicit `StandardCopyOption`s, `delete` which throws a descriptive exception, and `walk` / `list` / `find` inside try-with-resources because they hold OS handles. For crash-safe updates I write to a temp file in the same directory and `move` it with `ATOMIC_MOVE`."

- *Why prefer `Files.delete` over `File.delete`?* → It throws a specific exception explaining the failure; `File.delete` just returns `false`.
- *How do you delete a directory tree?* → `walk`, sort in reverse order, delete each; or `walkFileTree` deleting in `postVisitDirectory`.
- *Is `walk` depth-first or breadth-first?* → Depth-first; it includes the start directory.

Self-check

1. What does `Files.copy(dir, target)` do for a non-empty directory? → Creates an empty directory; contents are not copied.
2. What does `Files.mismatch` return for identical files? → `-1`.
3. Why sort the results of `Files.list` in a test? → The OS order is unspecified.

13.4 Reading and writing whole files

What it is

For everyday text files, `Files` has one-line methods. They default to **UTF-8** (the dominant text encoding, explained in §13.7; this was true for `Files` methods even before Java 18) and open/close the file for you.

Method	Reads/writes	Memory	Since
`readString(p[, cs])` / `writeString(p, text[, cs], opts...)`	whole file as one `String`	whole file	11
`readAllLines(p)` / `write(p, lines, opts...)`	`List<String>` of lines	whole file	7
`readAllBytes(p)` / `write(p, bytes, opts...)`	`byte[]`	whole file	7
`lines(p)`	lazy `Stream<String>` — **must be closed**	one line at a time	8
`newBufferedReader(p)` / `newBufferedWriter(p, opts...)`	`BufferedReader` / `BufferedWriter`	small buffer	7
`newInputStream(p)` / `newOutputStream(p, opts...)`	raw bytes	small buffer	7

Open options (`StandardOpenOption`) control what "write" means. The default for the write methods is `CREATE`, `TRUNCATE_EXISTING`, `WRITE` — create if missing, **otherwise wipe it first** (*truncate* = cut the file to zero length). Others: `APPEND` (add at the end), `CREATE_NEW` (fail if it exists — an atomic "create only if absent"), `READ`, `SYNC`/`DSYNC` (force every write to the storage device — `DSYNC` syncs only the contents, `SYNC` also the metadata), `DELETE_ON_CLOSE`.

`code/ch13/files/ReadWrite.java` (key parts):

```
Files.writeString(f, "buy milk\n");                              // create or truncate
Files.writeString(f, "call bob\n", StandardOpenOption.APPEND);
Files.write(f, List.of("fix bug #42", "ship v2"), StandardOpenOption.APPEND);
Files.readString(f);
List<String> lines = Files.readAllLines(f);
try (Stream<String> s = Files.lines(f)) {                         // lazy; MUST be closed
    long withB = s.filter(l -> l.contains("b")).count();
}
Files.writeString(f, "x", StandardOpenOption.CREATE_NEW);         // FileAlreadyExistsException

try (BufferedWriter w = Files.newBufferedWriter(csv, StandardCharsets.UTF_8)) {
    w.write("name,score"); w.newLine();
    w.write("ana,91");     w.newLine();
    w.write("raj,78");     w.newLine();
}                                                                 // close() flushes the buffer
try (BufferedReader r = Files.newBufferedReader(csv)) {
    r.readLine();                                                 // skip header
    String line;
    while ((line = r.readLine()) != null) total += Integer.parseInt(line.split(",")[1]);
}
```

Output:

```
readString:
buy milk
call bob
fix bug #42
ship v2
readAllLines size = 4 -> [buy milk, call bob, fix bug #42, ship v2]
lines containing 'b' = 3
bytes = 38
CREATE_NEW on existing -> FileAlreadyExistsException
total score = 169
```

- `Files.write(path, lines)` writes each line followed by the platform line separator (`\n` here). `readAllLines` strips line terminators.
- `readLine()` returns `null` at end of file — the classic `while ((line = r.readLine()) != null)` idiom.

Trade-offs

Style	Pros	Cons
`readString` / `readAllLines`	shortest code	loads the whole file; a 4 GB log → `OutOfMemoryError` (the JVM ran out of heap memory)
`lines()`	constant memory, stream operations	must remember try-with-resources; `UncheckedIOException` if a read fails mid-stream
`BufferedReader` loop	constant memory, full control, easy early exit	more code

Rule of thumb: whole-file methods for files you know are small (configs, test **fixtures** — fixed sample data used by tests — < a few MB); streaming for anything user-supplied or unbounded.

Common mistakes

- Forgetting that the default write **truncates** — appending needs `APPEND`.
- Writing a CSV "by hand" with `split(",")` for real data: quoted fields containing commas break it. Use a CSV **library** (reusable code someone else wrote that you add as a dependency — e.g. Apache Commons CSV, OpenCSV) for anything external.
- Writing important files in place. If the program crashes mid-write the file is corrupt. Write a temp file in the same directory, then `Files.move(tmp, target, ATOMIC_MOVE, REPLACE_EXISTING)`.

Self-check

1. Which option makes `writeString` fail if the file exists? → `CREATE_NEW`.
2. Why wrap `Files.lines` in try-with-resources? → It keeps the file open until the stream is closed.
3. What does `readLine()` return at end of file? → `null`.

TL;DR

`readString`/`writeString` for small text, `lines`/`newBufferedReader` for big text, and remember: writes truncate unless you say `APPEND`.

13.5 Byte streams and character streams

What it is

In `java.io` all I/O is built from four **abstract classes** (classes that can't be instantiated directly and exist to be extended):

	Input (read)	Output (write)	Unit
Byte streams	`InputStream`	`OutputStream`	`byte` (8-bit raw data: images, zip, protobuf — Google's compact binary data format — anything)
Character streams	`Reader`	`Writer`	`char` (16-bit UTF-16 text unit; see §13.7)

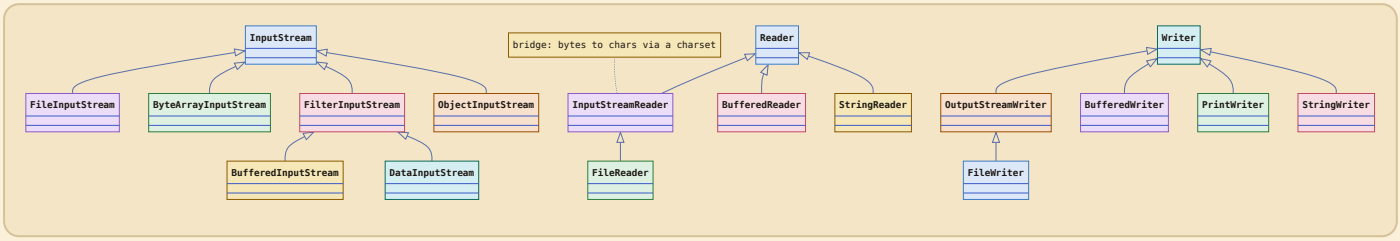
Byte streams move raw bytes. Character streams move text, and at the point where text meets bytes (a file, a socket) something must **encode** chars to bytes or **decode** bytes to chars using a **charset** (§13.7). The **bridge** classes are `InputStreamReader` (bytes → chars) and `OutputStreamWriter` (chars → bytes).

How it works: the decorator pattern

Concrete classes come in two kinds:

1. **Source/sink streams** that actually connect to something: `FileInputStream`, `ByteArrayInputStream`, a socket's stream, `FileReader`, `StringReader`, `StringWriter`.
2. **Filter (wrapper) streams** that take another stream in their constructor and add a feature: buffering (`BufferedInputStream`, `BufferedReader`), primitives (`DataInputStream`), objects (`ObjectInputStream`), compression (`GZIPInputStream`), formatting (`PrintStream`, `PrintWriter`).

You stack wrappers like nested boxes. Each `read`/`write` passes through the layers, and **closing the outermost wrapper closes the whole chain**. This is the **Decorator** pattern (wrapping an object in another object of the same type that adds behaviour, LLD Chapter 9).



(The output side mirrors this: `OutputStream` → `FileOutputStream`, `ByteArrayOutputStream`, `BufferedOutputStream`, `DataOutputStream`, `ObjectOutputStream`, `PrintStream`.)

The read contracts, which every subclass follows:

- `int read()` returns the next byte (0–255) or char (0–65535) **as an `int`**, or **`-1` at end of stream**. It returns `int` rather than `byte` precisely so `-1` can't be confused with a real value.
- `int read(byte[] buf)` fills *up to* `buf.length` bytes and returns **how many it actually read** (possibly fewer!) or `-1`. Always write `buf, 0, n`, never the whole array.
- Reads **block**: the **thread** (the flow of execution running your code) waits until data arrives, the stream ends, or an error occurs.

`code/ch13/streams/ByteAndCharStreams.java` (key parts):

```java
// 1) the classic copy loop
try (InputStream in = new FileInputStream(src.toFile());
     OutputStream out = new FileOutputStream(dst.toFile())) {
    byte[] buf = new byte[4];                  // tiny, to make the loop visible
    int n;
    while ((n = in.read(buf)) != -1) out.write(buf, 0, n);
}
// 2) Java 9: let the library do the loop
in.transferTo(out);
// 3) characters: specify the charset at the byte boundary
try (Writer w = new OutputStreamWriter(new FileOutputStream(text.toFile()), StandardCharsets.UTF_8)) {
    w.write("naïve café ☕");
}
// 4) decorators: text -> buffered -> encoded -> gzipped -> file
try (PrintWriter pw = new PrintWriter(new BufferedWriter(new OutputStreamWriter(
        new GZIPOutputStream(new FileOutputStream(gz.toFile())), StandardCharsets.UTF_8)))) {
    for (int i = 1; i <= 1000; i++) pw.printf("line %d: all work and no play%n", i);
}
// 5) in memory
ByteArrayOutputStream bos = new ByteArrayOutputStream();
new StringReader("abc").transferTo(sw);         // Reader.transferTo: Java 10
```

Output:

```
copied with 4 read() calls -> Hello, bytes!
transferTo copied 13 bytes
chars written = 12, bytes on disk = 16
chars read back = 12
gzip file 2453 bytes holds 1000 lines
ByteArrayOutputStream -> captured
StringReader -> StringWriter: abc
```

- 13 bytes through a 4-byte buffer = 4 reads (4+4+4+1); the last `write(buf, 0, 1)` writes only one byte.
- 12 characters became 16 bytes: `ï` and `é` take 2 bytes in UTF-8, `☕` takes 3.
- ~30 KB of repetitive text compressed to about 2.4 KB — and the code reading it back just adds `GZIPInputStream` to the chain.

Useful extras (Java 9–25)

Method	What it does	Since
`InputStream.readAllBytes()`, `readNBytes(n)`	read everything / exactly up to n bytes	9 / 11
`InputStream.transferTo(out)`, `Reader.transferTo(w)`	copy to the end	9 / 10
`InputStream.nullInputStream()`, `OutputStream.nullOutputStream()`	empty source / discard sink (tests)	11
`Reader.of(CharSequence)`	a `Reader` over a string	24
`Reader.readAllAsString()`, `readAllLines()`	read the rest of a reader	25

`mark`/`reset` (remember a position and go back) and `skip` also exist; only some streams support `mark` (check `markSupported()`) — details in OCP Chapter 14 §8a.

Common mistakes

- `FileReader` / `FileWriter` **without a charset** on Java 17 and earlier used the platform default (often Windows-1252 on Windows) — files written on one machine read wrongly on another. On Java 11+ pass the charset: `new FileReader(file, StandardCharsets.UTF_8)`.
- Ignoring the return value of `read(buf)` and writing the whole buffer → garbage at the end of the copy.
- Closing only the inner stream, or not closing at all — buffered data is lost and the file handle leaks. Declare the outermost wrapper in try-with-resources.
- Using `PrintWriter` / `PrintStream` for important output: they **swallow** `IOException`**s** (check `checkError()`), which is fine for the console but not for writing a file you need.

In the interview

> **30-second answer:** "`java.io` has byte streams, `InputStream` / `OutputStream`, and character streams, `Reader` / `Writer`, bridged by `InputStreamReader` / `OutputStreamWriter` with an explicit charset. Features are added by wrapping — buffering, data types, compression — which is the Decorator pattern. `read` returns -1 at end of stream, and I always close the outermost wrapper with try-with-resources."

- *Why does `read()` return `int`?* → To fit all 256 byte values plus `-1` as the end marker.
- *Byte or character stream for a PNG?* → Byte stream; it isn't text.

Self-check

1. Which class turns bytes into chars? → `InputStreamReader` (with a charset).
2. What does closing a `BufferedReader` wrapping an `InputStreamReader` do? → Closes the whole chain.
3. What's wrong with `out.write(buf)` in a copy loop? → Writes stale bytes on the last partial read; use `write(buf, 0, n)`.

13.6 Buffering

What it is

A **buffer** is a block of memory (8 KiB — 8 × 1024 bytes — by default for the `Buffered*` classes) that collects many small reads or writes into one large operation. Without it, each `write('x')` on a `FileOutputStream` becomes a **system call** — a request to the operating-system **kernel** (the core of the OS that controls hardware such as disks), which costs far more than copying a byte in memory. Analogy: carrying groceries from the car one item per trip versus filling a bag.

How it works

1. `BufferedOutputStream.write(b)` puts the byte in its internal array.
2. When the array is full, it writes the whole block to the wrapped stream in one call.
3. `flush()` forces out whatever is buffered now; `close()` flushes then closes.
4. On input, `BufferedInputStream` / `BufferedReader` read a large block at once and hand out pieces from memory; `BufferedReader` also adds `readLine()` and `lines()`.

`code/ch13/streams/BufferingDemo.java` (key parts):

```java
try (OutputStream out = new FileOutputStream(f1.toFile())) {
    for (int i = 0; i < 200_000; i++) out.write('x');          // one system call per byte
}
try (OutputStream out = new BufferedOutputStream(new FileOutputStream(f2.toFile()))) {
    for (int i = 0; i < 200_000; i++) out.write('x');          // one system call per 8 KiB
}
BufferedWriter w = new BufferedWriter(new FileWriter(lost.toFile()));
w.write("this sits in the buffer");
Files.size(lost);   // 0 !
w.flush();
Files.size(lost);   // 23
```

Output (timings are machine-dependent; the ratio is the lesson):

```
same size: true (200000 bytes)
unbuffered: 205 ms   [machine-dependent]
buffered:   2 ms   [machine-dependent]
buffered faster? true
before flush: 0 bytes on disk
after flush:  23 bytes on disk
```

Same 200,000 bytes, roughly 100× faster with a buffer on this machine. And until `flush()`/`close()`, written data lives only in the buffer: if the JVM crashes or you forget to close, it never reaches the file.

When not to use it

- When you already read/write big arrays (`read(byte[8192])`, `Files.readAllBytes`, `transferTo`) — a buffer adds only a copy.
- The `Files` convenience methods and `Files.newBufferedReader/Writer` already buffer.
- Interactive **protocols** (agreed message formats between two programs, e.g. over a network) where each message must go out immediately — you can still buffer, but call `flush()` after each message.

Common mistakes

- Forgetting `flush()` on a socket writer → the other side waits forever for a message sitting in your buffer.
- Wrapping twice (`new BufferedReader(Files.newBufferedReader(p))`) — harmless but pointless.
- Note: `flush()` hands data to the OS, not necessarily to the physical disk. For **durability** (data surviving a crash or power cut) after power loss use `FileChannel.force(true)` or the `SYNC`/`DSYNC` open options (databases do this).

TL;DR

Always buffer small reads/writes; `close()` flushes; `flush()` ≠ on disk.

13.7 Charsets and encodings

What it is

Files and networks carry **bytes**; Java `String`s hold **characters** (as UTF-16 `char` units). A **charset** (character encoding) is the rulebook that maps between them.

- **UTF-8** — variable length: 1 byte for ASCII (the basic 128 English letters, digits and symbols), 2 for most European accents, 3 for most other symbols (`€`, CJK — Chinese, Japanese, Korean

characters), 4 for emoji. The web's and Java's default.

- **UTF-16** — 2 or 4 bytes per character; what `String` uses internally (conceptually).
- **ISO-8859-1** (Latin-1) — exactly 1 byte per character, only 256 characters.
- **US-ASCII** — 7-bit, 128 characters.

Mojibake is the garbled text you get when bytes are decoded with a different charset from the one used to encode them (`cafÃ©`).

Why it matters: JEP 400 (Java 18)

A **JEP** (*JDK Enhancement Proposal*) is the numbered design document behind each new Java feature. Before Java 18, APIs without an explicit charset (`FileReader`, `new String(bytes)`, `String.getBytes()`, `Scanner(File)`) used the **platform default**, which depended on the OS and **locale** (the user's language/region settings) (UTF-8 on most Linux/macOS, Windows-1252 on many Windows machines). The same program could read a file correctly on a developer's Mac and wrongly on a Windows server. Since **Java 18 (JEP 400) the default charset is UTF-8 everywhere**. (The console's own encoding is separate and still follows the terminal: see `System.getProperty("stdout.encoding")`.) Being explicit — `StandardCharsets.UTF_8` — is still best practice, because it documents intent and keeps code correct if someone runs it with `-Dfile.encoding=COMPAT` (`-Dname=value` on the `java` command line sets a *system property*, §13.11).

`code/ch13/charsets/CharsetDemo.java` (key parts):

```
String s = "café €";
for (Charset cs : new Charset[]{UTF_8, UTF_16BE, ISO_8859_1, US_ASCII}) {
    byte[] b = s.getBytes(cs);                                  // encode
    System.out.printf("%-10s %2d bytes  %s%n", cs, b.length, hex.formatHex(b));
}
byte[] utf8 = "café".getBytes(UTF_8);
new String(utf8, ISO_8859_1);                                   // decode with the wrong charset
Files.writeString(latin, "café", ISO_8859_1);
Files.readString(latin);                                        // assumes UTF-8 -> throws
String e = "hi👋";  e.length(); e.codePoints().count();
```

Output:

```
default charset = UTF-8 (UTF-8 by default since Java 18, JEP 400)
UTF-8       9 bytes  63 61 66 c3 a9 20 e2 82 ac
UTF-16BE   12 bytes  00 63 00 61 00 66 00 e9 00 20 20 ac
ISO-8859-1  6 bytes  63 61 66 e9 20 3f
US-ASCII    6 bytes  63 61 66 3f 20 3f
decoded wrong  -> cafÃ©
decoded right  -> café
readString(latin1 file) -> MalformedInputException
with right charset -> café
length()=4 codePoints=3 utf8 bytes=6
```

What to notice:

- The same 6-character string is 9, 12 or 6 bytes depending on the charset.
- `3f` is `?`: `String.getBytes` **silently replaces** characters the charset can't represent (`€` in Latin-1; `é` and `€` in ASCII). Data is lost without an error. (To get an error instead, use a `CharsetEncoder` with `CodingErrorAction.REPORT`.)
- `Files.readString`/`readAllLines` are **strict**: invalid UTF-8 throws `MalformedInputException` rather than guessing — better than silent corruption. `BufferedReader` built with `InputStreamReader` is lenient and substitutes `�` (U+FFFD).

- `"hi👋".length()` is 4: the emoji is one **code point** but two `char`s (a *surrogate pair*). Use `codePoints()` to count what humans see as characters (more in Chapter 7).

Common mistakes

- Decoding a byte array that was cut in the middle of a multi-byte character (e.g. splitting network data at a fixed size) — decode after reassembling, or use a `Reader` which handles the boundary.
- A **BOM** (byte-order mark, `EF BB BF`) at the start of files from some Windows tools: Java's UTF-8 decoder keeps it as the character `\uFEFF`, so the first header of a CSV might not equal `"name"`. Strip it explicitly.
- Using `Properties.load(InputStream)`, which assumes ISO-8859-1 (§13.11).

In the interview

> **30-second answer:** "Strings are characters; files are bytes; a charset converts. Since Java 18 the default is UTF-8 on every platform, but I still pass `StandardCharsets.UTF_8` explicitly. Wrong charset gives mojibake; `getBytes` silently substitutes `?` for unmappable characters while `Files.readString` throws on malformed input."

- *How many bytes is `é` in UTF-8?* → 2.
- *Why can `String.length()` differ from the number of visible characters?* → It counts UTF-16 `char`s; characters outside the Basic Multilingual Plane (emoji) take two.

Self-check

1. What was JEP 400? → UTF-8 as the default charset, Java 18.
2. What does ISO-8859-1 do with `€`? → Can't represent it; `getBytes` writes `?`.
3. Which is strict on bad input: `Files.readString` or `InputStreamReader`? → `Files.readString`.

13.8 Console and standard streams

What it is

Every process starts with three **standard streams** connected by the OS: **standard input** (`System.in`, an `InputStream` — usually the keyboard), **standard output** (`System.out`, a `PrintStream`) and **standard error** (`System.err`, a `PrintStream` for diagnostics). **Shells** (the command-line programs, like bash or zsh, that you type commands into) can **redirect** them (point them at files instead of the screen/keyboard) independently: `java App < input.txt > out.txt 2> errors.txt`.

Ways to read input:

Tool	Best for	Notes
`Scanner(System.in)`	small interactive programs, exercises	parses tokens (`nextInt`, `nextDouble`, `nextLine`); slow for big input
`BufferedReader(new InputStreamReader(System.in))`	large inputs (competitive programming, piped data — output of one command fed straight into another)	line-based; parse with `split` / `Integer.parseInt`
`System.console()`	real terminals; passwords	`readPassword()` doesn't echo and returns `char[]` (can be wiped)
`IO.readln(prompt)` / `IO.println(x)`	beginner programs	`java.lang.IO`, **Java 25** (JEP 512), usable without imports

`code/ch13/console/ConsoleInput.java` simulates typed input with an in-memory stream so the output is repeatable:

```
String typed = "Ada\n36\n3 4 5\n";
Scanner sc = new Scanner(new ByteArrayInputStream(typed.getBytes(UTF_8)));  // real code: new Scanner(System.in)
String name = sc.nextLine();
int age = sc.nextInt();
while (sc.hasNextInt()) sum += sc.nextInt();

BufferedReader br = new BufferedReader(new StringReader(typed));  // real code: new InputStreamReader(System.in)
String first = br.readLine();
int parsedAge = Integer.parseInt(br.readLine().trim());
for (String tok : br.readLine().split("\\s+")) total += Integer.parseInt(tok);

Scanner trap = new Scanner("42\nhello\n");
int n = trap.nextInt();
String rest = trap.nextLine();           // "" - the rest of the line after 42!

Console c = System.console();
c != null && c.isTerminal();             // isTerminal(): Java 22
IO.println("IO.println works without System.out");             // Java 25
```

Output (the stderr line may appear in a different position when both streams go to one terminal — they are separate streams):

```
Scanner: name=Ada age=36 sum=12
BufferedReader: name=Ada age=36 sum=12
trap: n=42 nextLine=[] then [hello]
stdout and stderr are separate streams (redirect with > and 2>)
this line went to stderr
System.console() available? false
IO.println works without System.out
```

- **The `nextInt`/`nextLine` trap**: `nextInt()` consumes `42` but leaves the newline; the following `nextLine()` returns the empty remainder of that line. Fix: call an extra `nextLine()` after `nextInt()`, or read whole lines and parse them.
- `System.console()` returns `null` when input or output is redirected (IDEs, pipes, CI — *continuous integration*, servers that automatically build and test every code change) — always null-check it. Java 22–24 used a JLine-based (JLine is a library for rich terminal input) console by default

that could return a `Console` even then; Java 25 reverted to the old behaviour. `isTerminal()` (22) tells you whether a returned console is really interactive.

Common mistakes

- Closing a `Scanner` wrapping `System.in` in a helper method — that closes `System.in` for the rest of the program.
- Using `System.out.println` for logging in applications — use a logging framework (a library that writes timestamped, levelled, configurable log messages) (Chapter 18).
- Reading passwords with `Scanner` into a `String` — use `Console.readPassword()`.

Self-check

1. Which stream should error messages go to? → `System.err`.
2. Why is `BufferedReader` preferred for 10^6 numbers? → `Scanner` uses **regex** (regular expression: a pattern language for matching text) parsing and is much slower.
3. What does `IO.println` need to be imported? → Nothing; `java.lang.IO` is imported automatically (Java 25).

13.9 Binary data, channels and buffers

Primitive values: `DataOutputStream` / `DataInputStream`

To store numbers compactly (not as text), `DataOutputStream` writes Java **primitives** (`int`, `double`, `boolean` …) in a fixed **big-endian** (most significant byte first) binary format; `DataInputStream` reads them back. **You must read in exactly the same order and types you wrote** — the file has no labels.

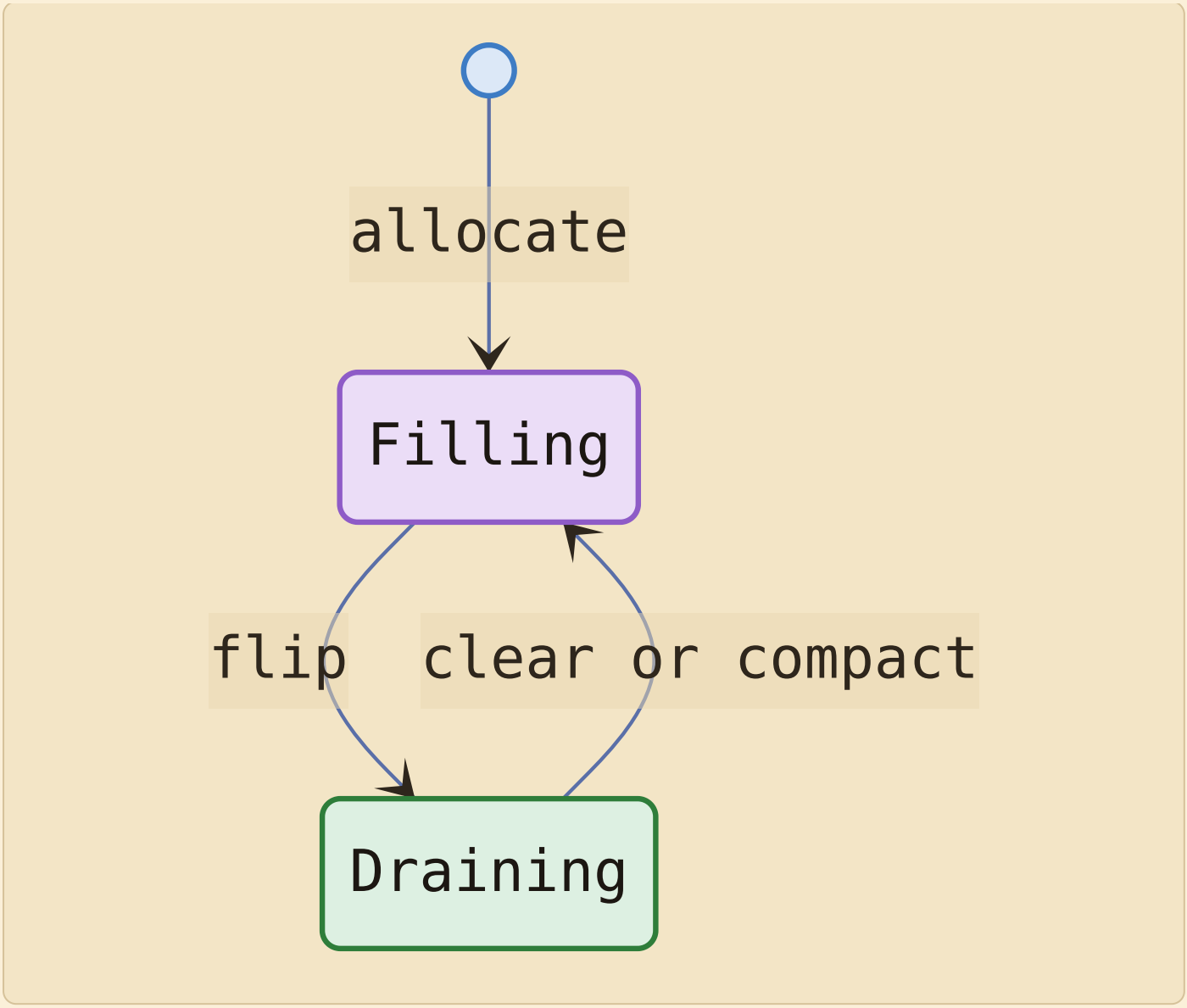
`code/ch13/binary/DataAndChannels.java` (key parts):

```
try (DataOutputStream out = new DataOutputStream(new BufferedOutputStream(Files.newOutputStream(f))))
{
    out.writeInt(2025);  out.writeDouble(3.5);  out.writeBoolean(true);  out.writeUTF("ok");
}
try (DataInputStream in = new DataInputStream(new BufferedInputStream(Files.newInputStream(f)))) {
    in.readInt(); in.readDouble(); in.readBoolean(); in.readUTF();
}
```

Channels and `ByteBuffer`

NIO models I/O differently from streams:

- A **channel** (`FileChannel`, `SocketChannel`) is a connection that can read *into* and write *from* a buffer, often in both directions, and supports **positioned** (random-access) reads.
- A `ByteBuffer` is a fixed-capacity block of bytes with three cursors: **position** (next index to use), **limit** (end of valid data), **capacity**. You *fill* it (channel read), call `flip()` to switch to *draining* (limit = position, position = 0), consume it, then `clear()` to fill again (position = 0, limit = capacity) — or `compact()`, which keeps any unread bytes at the front.




```
try (FileChannel ch = FileChannel.open(g, READ)) {
    ByteBuffer buf = ByteBuffer.allocate(8);
    while (ch.read(buf) != -1) {         // fill
        buf.flip();                      // switch to draining
        sb.append(UTF_8.decode(buf));
        buf.clear();                     // ready to fill again
    }
}
ch.read(word, 13);                       // positioned read at byte offset 13
```

Output:

```
file size = 17 bytes (4 + 8 + 1 + 2 + 2)
read back: 2025, 3.5, true, ok
channel wrote 20 bytes, position now 20
channel read: channels and buffers
bytes 13..15 = buf
```

(`writeUTF("ok")` = a 2-byte length prefix + 2 bytes. Decoding an 8-byte UTF-8 chunk independently is only safe here because the text is ASCII — with multi-byte characters use a `CharsetDecoder` or a `Reader`.)

What else NIO offers (overview)

Feature	What it's for
`FileChannel.map(...)` → `MappedByteBuffer`	**memory-mapped files**: the OS maps the file into memory; fast random access to huge files (databases, search indexes)
`FileChannel.transferTo/transferFrom`	**zero-copy** (the OS moves the data without copying it through your program's memory) file-to-socket transfer (web servers sending static files)
`FileChannel.lock()`	OS-level file locks between processes
`Selector` + non-blocking `SocketChannel`	one thread watching thousands of connections — the basis of Netty (a popular high-performance networking library); far less needed now that **virtual threads** (lightweight JVM-managed threads) make blocking I/O cheap (Chapter 15)
`AsynchronousFileChannel`	**callback** (a function you hand over to be called when the work finishes) / `Future`-based (a handle to a result that will be ready later) file I/O
`ByteBuffer.allocateDirect`	memory outside the **heap** (the JVM-managed memory area where objects live and the garbage collector cleans up) for **native** I/O (I/O done directly by OS / C code) (the newer Foreign Function & Memory API, final in Java 22, is the modern route to off-heap memory)
`RandomAccessFile` (java.io)	the older way to seek and read/write at positions

Self-check

1. What does `flip()` do? → Sets limit to the current position and position to 0, so you can read what you just wrote into the buffer.
2. Can a `DataInputStream` tell you what type is next? → No; the reader must know the layout.

13.10 Serialization (and why to avoid it)

What it is

Java serialization converts an **object graph** (an object and everything it references) into a byte stream with `ObjectOutputStream.writeObject`, and rebuilds it with `ObjectInputStream.readObject`. A class opts in by implementing the **marker interface** `java.io.Serializable` (an interface with no methods that only "tags" a class).

Rules:

1. Every non-`transient`, non-`static` field is written; referenced objects must also be `Serializable`, or `NotSerializableException` is thrown.
2. `transient` fields are skipped and come back as default values (`null`, `0`, `false`).
3. On deserialization of a normal class, **its constructor does not run**. Java allocates the object and fills fields directly; only the no-arg constructor of the first *non-serializable* superclass runs. Validation in constructors is therefore bypassed.
4. **Records** are the exception: they are rebuilt **through the canonical constructor** (the record's main constructor taking every component in order), so validation runs (Java 16).
5. `serialVersionUID` is a version stamp. If you don't declare one, the JVM computes it from the class structure; almost any change (adding a method) changes it, and old data then fails with `InvalidClassException`. Declare it explicitly (`@Serial private static final long serialVersionUID =`

`1L;` — `@Serial` is Java 14) if you serialize at all. (*Deserialization* is the reverse step: bytes back into objects.)

`code/ch13/serial/SerializationDemo.java` (key parts):

```
static class Account implements Serializable {
    @Serial private static final long serialVersionUID = 1L;
    private final String owner;
    private final long balanceCents;
    private transient String sessionToken;                    // skipped
    Account(...) { ...; System.out.println("  constructor ran for " + owner); }
}
record Point(int x, int y) implements Serializable {
    Point { if (x < 0 || y < 0) throw new IllegalArgumentException("negative"); ... }
}

ObjectInputFilter allow = ObjectInputFilter.Config.createFilter(
        "ch13.serial.SerializationDemo$Account;ch13.serial.SerializationDemo$Point;!*");
in.setObjectInputFilter(allow);                               // before readObject()
```

Output:

```
create:
  constructor ran for ana
serialized to 116 bytes
deserialize (note: NO constructor call for a normal class):
  Account[ana, 1250, token=null]  same object? false
record - create:
  canonical constructor ran for (1,2)
record - deserialize (the canonical constructor runs again):
  canonical constructor ran for (1,2)
  Point[x=1, y=2]
  canonical constructor ran for (1,2)
filtered, allowed class: Point[x=1, y=2]
filtered, ArrayList -> InvalidClassException: filter status: REJECTED
```

- The transient token came back `null`; the account is a new object with equal state.
- The record's canonical constructor ran on every deserialization (including the filtered one) — so an invalid byte stream can't create an invalid `Point`.
- The allow-list filter (`!*` = reject everything else) blocked an `ArrayList` before any of its code ran.

Why to avoid it

Effective Java (Item 85) is blunt: there is no reason to use Java serialization in new systems.

1. **Security.** `readObject` can **instantiate** (create an object of) any serializable class on the **classpath** (the list of folders and JAR files the JVM loads classes from) and run its deserialization methods. Attackers chain such "gadget" classes (existing harmless-looking classes whose side effects can be combined into an attack) to execute code — many real **remote-code-execution** (RCE: an attacker running their own code on your server) vulnerabilities in application servers and libraries came from deserializing untrusted bytes. Filters (`ObjectInputFilter`, Java 9, JEP 290; JVM-wide filter factories, Java 17, JEP 415; or the `jdk.serialFilter` system property) reduce the risk but must be configured correctly.
2. **Hidden constructor.** Deserialization bypasses constructors and invariants (except for records).
3. **Fragile versioning** and a format only Java can read.
4. **Freezes your implementation**: private fields become part of the public **wire format** (the exact byte layout sent over a network or stored on disk).

Use instead: JSON (§13.12) for human-readable data and APIs, Protocol Buffers/Avro (compact binary formats with a schema, readable from many languages) for compact cross-language data, a database for **persistence** (keeping data after the program stops). If you're stuck with serialization (legacy RMI — Remote Method Invocation, Java's old way to call methods on another machine — some caches), serialize records, set an allow-list filter, and never deserialize data from untrusted sources.

In the interview

30-second answer: "Serialization turns an object graph into bytes via `Serializable`; `transient` fields are skipped, `serialVersionUID` guards compatibility, and normal classes are rebuilt without calling their constructor. I avoid it for new code because deserializing untrusted data can lead to remote code execution and it's brittle; I use JSON or protobuf. If I must, I use records and an `ObjectInputFilter` allow-list."

- *What happens to a `static` field?* → Not serialized; it belongs to the class.
- *Superclass not `Serializable`?* → Its no-arg constructor runs on deserialization and its fields are not restored (and it must have an accessible no-arg constructor).

Self-check

1. Does a normal `Serializable` class's constructor run on `readObject`? → No.
2. And a record's? → Yes, the canonical constructor.
3. What exception when a referenced object isn't serializable? → `NotSerializableException`.

13.11 Properties files

What it is

A **properties file** is plain text of `key=value` lines — the traditional Java configuration format (Spring Boot's — Spring Boot being the most popular Java framework for building services — `application.properties`, `log4j2.properties`, `gradle.properties`). `java.util.Properties` loads and stores them.

Format rules: `#` or `!` start a comment; the separator can be `=`, `:` or whitespace, with spaces around it ignored; a trailing `\` continues the value on the next line (leading whitespace of the next line is dropped); values are **always strings**.

`code/ch13/props/PropertiesDemo.java` (key parts):

```
Files.writeString(file, """
        # database settings
        db.url = jdbc:postgresql://localhost:5432/shop
        db.pool.size=10
        ! bang also starts a comment
        greeting=Hello, \\
                world
        city=Zürich
        """);
Properties p = new Properties();
try (Reader r = Files.newBufferedReader(file, StandardCharsets.UTF_8)) { p.load(r); }
p.getProperty("db.url");
Integer.parseInt(p.getProperty("db.pool.size"));
p.getProperty("timeout", "30");                  // default when absent
p.setProperty("timeout", "45");
try (Writer w = Files.newBufferedWriter(out, UTF_8)) { p.store(w, "saved by demo"); }
```

Output:

```
keys        = [city, db.pool.size, db.url, greeting]
db.url      = jdbc:postgresql://localhost:5432/shop
pool+1      = 11   (values are always Strings)
greeting    = Hello, world
city        = Zürich
timeout     = 30   (default used)
saved 5 key lines
system property java.version starts with 2? true
```

(`\\` in the Java text block produces a single `\` in the file.)

Common mistakes

- `load(InputStream)` decodes as **ISO-8859-1**, so `Zürich` saved as UTF-8 becomes `ZÃ¼rich`. Use `load(Reader)` with UTF-8. (Resource bundles, `PropertyResourceBundle`, read UTF-8 since Java 9.)
- `Properties` extends `Hashtable<Object,Object>` (an old, synchronised map class; a historical design mistake); use only `getProperty` / `setProperty` / `stringPropertyNames`, never `put` / `get` with non-strings.
- **Committing** (saving into version control such as Git) secrets (passwords, **API keys** — secret tokens that identify your program to a service) in properties files — use **environment variables** (named values the OS passes to each process) or a **secret manager** (a service that stores and hands out credentials securely).
- **System properties** (`System.getProperty("user.home")`, set with `-Dkey=value`) are a separate, JVM-wide `Properties` object; **environment variables** come from `System.getenv("HOME")`.

For locale-specific messages with `ResourceBundle`, see OCP Chapter 11.

13.12 Working with JSON

What it is

JSON (JavaScript Object Notation: nested `{"key": value}` objects and `[ … ]` lists in plain text) is the text format of web APIs and most modern config/data files. As of Java 25/26 the JDK has **no public JSON API** — you use a library. The two common ones are **Jackson** (the default in Spring Boot) and **Gson**; JSON-B/Yasson is the Jakarta EE (the enterprise Java specifications, formerly Java EE) standard. Jackson 3 (October 2025, default in Spring Boot 4) moved to `tools.jackson.*` packages and prefers building an immutable `JsonMapper` via `JsonMapper.builder()`; the calls below are otherwise the same.

How it works (Jackson 2.x — not run: needs the `jackson-databind` dependency)

A **dependency** is an external library your project declares and the build tool downloads for you.

```
// (Jackson - not run)
record Order(String id, List<String> items, double total) {}

ObjectMapper mapper = new ObjectMapper();                  // thread-safe once configured; reuse one
String json = mapper.writeValueAsString(new Order("A1", List.of("pen", "ink"), 9.5));
// {"id":"A1","items":["pen","ink"],"total":9.5}
Order back = mapper.readValue(json, Order.class);          // records supported since Jackson 2.12
JsonNode tree = mapper.readTree(json);                     // untyped tree when there's no class
tree.get("items").get(0).asText();                         // "pen"
List<Order> orders = mapper.readValue(arrayJson, new TypeReference<List<Order>>() {});  // generics survive erasure
```

- **Data binding** (JSON ↔ your classes; records fit perfectly) for known shapes; the **tree model** (`JsonNode`) for dynamic or partial data; **streaming** (`JsonParser`) for gigantic files.
- `TypeReference` works around **type erasure** (generic type arguments like `<Order>` are removed after compilation, so they're unknown at run time; Chapter 9).
- Money: use `BigDecimal`, not `double` (Chapter 7).
- Adding the dependency with Maven/Gradle: Chapter 18.

Common mistakes

Hand-building JSON with string concatenation (quotes and escapes break it; it's an **injection** risk — attacker-controlled text changing the structure of what you build), creating a new `ObjectMapper` per request (it's expensive — share one), and silently ignoring unknown fields vs failing on them without deciding which you want (`FAIL_ON_UNKNOWN_PROPERTIES`).

13.13 Best practices and common mistakes (checklist)

1. **`Path` + `Files`** for files; `java.io.File` only for legacy APIs.
2. **try-with-resources** for every stream, reader, writer, channel, and for `Files.lines/list/walk/find`.
3. **Always name the charset** (`StandardCharsets.UTF_8`), even though it's the default since Java 18.
4. **Buffer** small reads/writes; remember `close()` flushes and `flush()` isn't a disk sync.
5. **Stream large inputs**; load whole files only when they're known to be small.
6. **Don't check-then-act** (`exists` then `create`); do the operation and handle `FileAlreadyExistsException` / `NoSuchFileException`.
7. **Write atomically**: temp file in the same directory + `move(..., ATOMIC_MOVE)`.
8. **Validate user-supplied paths**: `normalize()` then `startsWith(baseDir)`.
9. **Don't swallow `IOException`**: it carries the reason (disk full, permission denied). Wrap in `UncheckedIOException` inside lambdas (Chapter 8 §8.8).
10. **No Java serialization of untrusted data**; prefer JSON/protobuf; if unavoidable, records + filters.
11. **Tests**: use `@TempDir` (JUnit Jupiter, the current version of Java's standard test framework) or `Files.createTempDirectory`, and in-memory streams (`ByteArrayInputStream`, `StringReader`) to test parsing without files.
12. **Blocking I/O is fine on virtual threads** — thousands of **concurrent** (in progress at the same time) file/socket operations no longer need **async** APIs (ones that return immediately and deliver the result later) (Chapter 15 §15.4).

In the interview (chapter-level)

- *How would you count lines in a 50 GB log file?* → `Files.lines(p)` in try-with-resources (or a `BufferedReader` loop) — constant memory; never `readAllLines`.

- *How do you make a config update crash-safe?* → Write to a temp file, `force` if durability matters, then `ATOMIC_MOVE` over the old file.
- *NIO vs IO?* → `java.io` = blocking streams; NIO = channels/buffers/selectors (non-blocking sockets, memory mapping); NIO.2 = `Path`/`Files` file-system API. Most file code today is NIO.2 + `java.io` readers.

Chapter TL;DR

Topic	Remember
APIs	`java.io` streams (1.0), NIO channels/buffers (1.4), NIO.2 `Path`/`Files` (7) — use NIO.2 for files
`Path`	Just a name; `resolve` appends (absolute arg wins), `relativize` gives the route, `normalize` is textual
`Files` ops	Specific exceptions; `copy`/`move` need `REPLACE_EXISTING`; delete trees children-first; close `list`/`walk`/`find`/`lines`
Whole files	`readString`/`writeString` (11) for small files; writes truncate unless `APPEND`; `CREATE_NEW` = atomic create
Streams	Byte (`InputStream`/`OutputStream`) vs char (`Reader`/`Writer`); bridge with a charset; decorators stack; `read` → `-1` at end
Buffering	~100× fewer system calls; `close()` flushes; unflushed data is lost
Charsets	UTF-8 default since Java 18 (JEP 400); still pass it explicitly; wrong charset = mojibake; `getBytes` substitutes `?`
Console	`Scanner` (easy), `BufferedReader` (fast), `Console` (passwords, `isTerminal` 22), `IO` (25); beware `nextInt`+`nextLine`
Binary/NIO	`Data*Stream` for primitives in a fixed order; `ByteBuffer` fill → `flip` → drain → `clear`; memory mapping for huge files
Serialization	`Serializable`, `transient`, `serialVersionUID`; constructors skipped (records excepted); security risk — avoid, or filter
Properties	`load(Reader)` with UTF-8; values are strings; `getProperty(key, default)`
JSON	Needs a library (Jackson/Gson); bind to records

Where next: doing many I/O operations at once — threads and thread pools in Chapter 14 — Concurrency I, then virtual threads, which make blocking I/O scale, in Chapter 15. Exam traps: OCP Chapter 14 — I/O. Python comparison (`open`, `pathlib`): Python Chapter 9.